Selection with if and else

CSC103 Programming Fundamentals / Lecture 08

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The problem for this lecture

Classify an integer as negative, zero or positive. Draw a decision outline, then write an if-else chain.

Check the negative case first.

Learning objectives

- Implement one-way and two-way selection
- Build an ordered multi-branch decision
- Test boundaries and nested conditions

CLO-2
The if statement
Two-way selection
Multiple selections
Nesting and compound statements

The if statement

- An if condition must be Boolean.
- The body executes only when the condition is true.
- Braces clearly group all statements in the branch.

Example

```java
if (balance < 0) {
  System.out.println("Overdrawn");
}
```

Two-way selection

- An if-else chooses exactly one branch.
- Statements after the structure execute after either branch completes normally.

Example

```
if (marks >= 50) {
  System.out.println("Pass");
} else {
  System.out.println("Fail");
}
```

Multiple selections

- An else-if chain selects the first matching branch.
- Order overlapping thresholds from most restrictive to least restrictive.
- Independent if statements can execute several bodies.

Example

```
if (score >= 80) {
  band = "High";
} else if (score >= 50) {
  band = "Middle";
} else { band = "Low"; }
```

Nesting and compound statements

- A nested if expresses a decision inside another decision.
- An else attaches to the nearest unmatched if.
- Braces prevent misleading indentation.

Example

```java
if (registered) {
  if (paid) {
    System.out.println("Admit");
  }
}
```

Branch order determines the selected result

- An else-if chain stops testing after its first matching condition.
- Overlapping ranges need an order that gives the most specific case priority.
- A final else covers every case not selected earlier.

Example

```
If >=50 comes before >=80, a score of 90
selects the >=50 branch immediately.
```

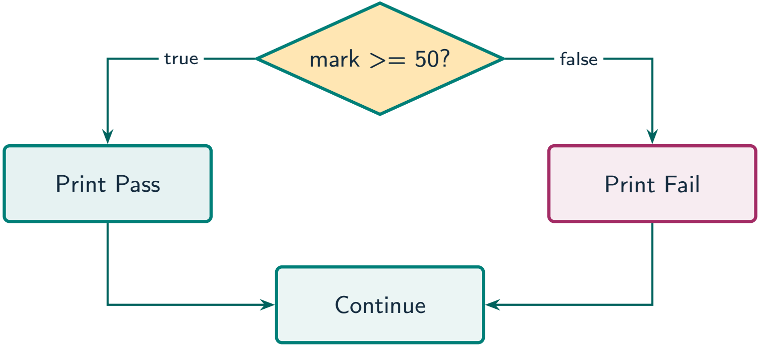
Independent tests and exclusive alternatives

- Separate if statements can execute more than one body.
- An if-else chain expresses mutually exclusive outcomes.
- For sign classification, exactly one of negative, zero and positive is appropriate.

Example

```
if (n < 0) ...
else if (n == 0) ...
else ...
```

A decision selects one path



What to notice

Exactly one branch executes; both normal paths rejoin after the selection.

Key distinctions

Input n	n < 0	n == 0 if reached	Chosen result
-1	true	Not evaluated	Negative
0	false	true	Zero
1	false	false	Positive

These distinctions determine which operation or control structure fits the problem.

First example: execution

```java
int score = 80;
String band;
if (score >= 80) { band = "High"; }
else if (score >= 50) { band = "Middle"; }
else { band = "Low"; }
System.out.println(band);
```

First example: result

High
The first condition is true.
The remaining branches are skipped.

Multiple selections
Nesting and compound statements

Guided practice

Classify an integer as negative, zero or positive. Draw a decision outline, then write an if-else chain.

Attempt the solution before continuing.
Use the definitions and examples from this lecture.

Solution strategy

- Check the negative case first.
- If the number is not negative, test equality to zero.
- Every remaining integer is positive.

The next program uses fixed inputs so its result can be reproduced.

Complete solution

```java
public class Solution08 {
  public static void main(String[] args) throws Exception {
    int n = 0;
    if (n < 0) {
      System.out.println("Negative");
    } else if (n == 0) {
      System.out.println("Zero");
    } else {
      System.out.println("Positive");
    }
  }
}
```

Execution trace

Execution point	Condition or action	Result
First test	$0 < 0$	false
Second test	$0 == 0$	true
Selected body	Print Zero	Zero
Remaining branch	Skip else	No extra output

Follow the changing state in execution order.

Solution result

Zero

Why this result follows

Every remaining integer is positive.

A common incorrect approach

```
if (marks >= 50);
{ System.out.println("Pass"); }
```

Example

The semicolon creates an empty `if` body. Remove it so that the block belongs to the `if`.

Boundary and failure checks

Check the stated input assumptions.
Write a shipping-charge decision for three weight ranges.
State your rules and test each boundary.

if (n < 0) ... else if (n == 0) ... else ...

Test -1, 0 and 1. Exactly one label must appear.

Check your understanding

How many branches can an else-if chain select? Why test 49, 50 and 51?

Write a prediction and explain the rule that supports it.

Answer and explanation

At most one. Those cases exercise both sides of the 50 boundary and equality itself.

The semicolon creates an empty if body. Remove it so that the block belongs to the if.

Compare cases and predict outcomes

Mark	Condition	Printed result
49	false	Fail
50	true	Pass
51	true	Pass

Discuss

Explain each expected result before running the example. Which case would reveal a wrong assumption?

Apply the idea

Independent practice

Write a grade band selection: 80 or above is High, 50 through 79 is Pass, and below 50 is Fail. Explain the order of tests.

1. Work through the input by hand.
2. Write or correct the program.
3. Justify the expected result.

Solution and reasoning

```java
if (mark >= 80) {
  System.out.println("High");
} else if (mark >= 50) {
  System.out.println("Pass");
} else {
  System.out.println("Fail");
}
```

- The narrower, higher threshold must be tested first.
- After mark >= 80 is false, mark >= 50 means 50 through 79.
- The example assumes a valid mark; range validation is a separate step.

Lecture summary

- one-way and two-way selection
- an ordered multi-branch decision
- boundaries and nested conditions

- Check the negative case first.
- If the number is not negative, test equality to zero.
- Every remaining integer is positive.

After-class practice

Write a shipping-charge decision for three weight ranges. State your rules and test each boundary.

Syllabus reading
Deitel Ch 7 (as listed)
Next lecture: Short-circuiting and switch